

MINGI KANG

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Last updated: April 28, 2026

RESEARCH INTEREST

My research interests focus on understanding chemical phenomena at the microscopic level through computational chemistry methodologies and accelerating these investigations using AI. Based on this fundamental understanding, I aim to explore and design novel materials and catalysts.

SKILLS

Computational Chemistry

- Ab initio calculation: Psi4, PySCF, GAMESS-US, Gaussian, ORCA, CP2K etc.
- Molecular Dynamics: OpenMM, ASE
- Modeling & Simulation: scikit-learn, pytorch, Training ML potentials

Scientific Skills

- HPC server & scheduling e.g. PBS, SGE
- Dataset handling (HDF5, SQL)
- Data visualization

Development Skills

- Workflow automation
- Python packaging & deployment
- Git, Cloud service, API

RESEARCH EXPERIENCE

Bachelor's Thesis

Lab of Ultrafast Spectroscopy, Korea University (Sejong)

March 2025 - Present

Prof. Jae Yoon Shin

- Investigated dynamic behavior of ionic liquids in porous filters (Polyethersulfone and Anodisc)
- Performed molecular dynamics calculations using force fields and fine-tuned ML potentials
- Built simulation systems and analyzed dynamic trajectories to determine diffusion coefficients

CURT Research Program

Laboratory of Inorganic Chemistry, Korea University (Sejong)

July 2025 - Present

Prof. Ho-Jin Son

- Studied reaction mechanisms of homogeneous transition metal catalysts using DFT and MLP methods
- Computed Gibbs energy profiles through various levels of DFT
- Developed automated job submission scripts and established workflows for Gibbs energy profile calculation

Winter Internship

The Meta Lab, KENTECH

January 2025 - February 2025

Prof. Geun Ho Gu

- Fine-tuned universal MLP models for heterogeneous catalyst systems
- Reproduced surface adsorption energies using MLP and compared results with DFT

Project Semester

Lab of Ultrafast Spectroscopy, Korea University (Sejong)

July 2024 - December 2024

Prof. Jae Yoon Shin

- Collected and preprocessed organic molecule dataset (10K samples) using the PubChem API
- Performed large-scale ab initio calculations (50K) and built normal mode perturbed molecular dataset
- Fine-tuned ANI Machine Learning Potentials (MLP) to improve accuracy in transition state regions
- Conducted MLP-based molecular dynamics simulations of carbon polymerizations and acetylene annulation reactions
- Developed Python package for MLP-based distortion interaction analysis calculations

PUBLICATIONS

H.S. Kim[†], Y.Y. Kim[†], H. Na[†], Y.-H. Seo, S.J. Na, S. Lee, **M. Kang**, J. Jang, D. Kim, Y. Ju, S.-M. Yoo, J.Y. Park, B.J. Kang, G.H. Gu, T.J. Shin, J. Lee^{*}, N.J. Jeon^{*}, E.H. Jung^{*}. Amorphous π -Conjugated Passivator Facilitates Widening Process Window for Reliable Perovskite Solar Modules. *Adv. Energy Mater.*, **Accepted (2026)**.

EDUCATION

KENTECH

Integrated M.S./Ph.D. Program, Department of Energy Engineering

March 2026 - Present

Prof. Geunho Gu

Korea University (Sejong) | GPA: 4.12 / 4.5

March 2019 - February 2026

B.S. in Advanced Material Chemistry

SELECTED PROJECTS

Distortion Interaction Analysis

- Implemented ASE interfaces for distortion interaction analysis calculations
- Enabled flexible computations and streamlined computational workflows

Molecule Aligner

- Developed a Python package for merging, interpolating, and aligning molecular trajectories
- Enabled construction of smooth, multi-step reaction pathways from mixed molecular inputs

Molecular Visualizer

- Developing molecular visualization tools including ASE-native viewer aseview and OverlayMol
- Supports visualization of molecular normal modes, overlay diagrams, and animations with publication-ready quality

ASE Community Code Development

- Implemented ASE calculator interfaces for g-xTB, PySCF, MLatom, and XequiNet
- Developed optimizer wrappers for asemd and geomeTRIC to enhance integration of ASE-centered workflows

SCHOLARSHIPS

Chi-Woo Lee Scholarship

- Korea University, 2025, KRW 5,000,000

Academic Excellence Scholarship

- Korea University, 2025, KRW 2,350,000

AWARDS & HONORS

Excellent Research Report Award

- S-CURT Research Program, Korea University, 2025

RESEARCH GRANTS

Project Semester Research Grant

- Korea University, 2024, KRW 1,500,000

ADDITIONAL ACTIVITY

Data Science Bootcamp

February 2024 - July 2024

- Completed a 6-month intensive course on data science and machine learning

MILITARY SERVICE

Republic of Korea Army (Sergeant · Mandatory Military Service)

April 2021 - October 2022

- Completed mandatory military service.